Quartic Oscillator at One Loop

Local adiabatic dynamics and causal nonlocal memory

Scope and results

For $S[q] = \int \mathrm{d}t\,[m\dot{q}^2/2 - m\omega^2 q^2/2 - \lambda q^4/24]$, this note derives two complementary one-loop equations. Equation (A) is local, second adiabatic order, and nonperturbative in the background amplitude. Equation (B) is causal and retarded, but first Born in $\delta\Omega^2 = \lambda Q^2/(2m)$. Their coefficients agree in the common weak-background, low-frequency, adiabatically prepared regime. Every remainder and state assumption used below is stated explicitly.

1 Model, assumptions, and power counting

The Euclidean action and local fluctuation frequency are

$$S_E[q] = \int \mathrm{d}\tau \left[\frac{m}{2}\dot{q}^2 + \frac{m\omega^2}{2}q^2 + \frac{\lambda}{24}q^4\right], \qquad \Omega^2(Q) = \omega^2 + \frac{\lambda Q^2}{2m}. \tag{1}$$

We assume $m > 0$, $\omega > 0$, $\lambda \geq 0$, and $\Omega \geq \Omega_{\min} > 0$. Euclidean time is the zero-temperature infinite line; integrations by parts are legitimate, and the background and reference determinants use identical boundary conditions. "One loop" means terms through $O(\hbar)$; a self-consistent solution of a truncated equation may resum uncontrolled higher powers of $\hbar$, so strict loop accuracy requires perturbative solution or order reduction.

2 Euclidean one-loop action

2.1 Background expansion and determinant

Set $q = Q + \eta$. Using $\int m\dot{Q}\dot{\eta}\,\mathrm{d}\tau = -\int m\ddot{Q}\eta\,\mathrm{d}\tau$ and the binomial expansion,

$$S_E[Q+\eta] = S_E[Q] + \int \mathrm{d}\tau\,\eta E_Q + \frac{1}{2}\int \mathrm{d}\tau\,\eta O_Q \eta + \int \mathrm{d}\tau\left(\frac{\lambda}{6}Q\eta^3 + \frac{\lambda}{24}\eta^4\right), \tag{2}$$

$$E_Q = -m\ddot{Q} + m\omega^2 Q + \frac{\lambda}{6}Q^3, \qquad O_Q = m\left[-\partial_\tau^2 + \Omega^2(\tau)\right]. \tag{3}$$

The effective-action source cancels the linear term. Cubic and quartic fluctuation vertices start at two loops, hence

$$\Gamma_E[Q] = S_E[Q] + \frac{\hbar}{2}\operatorname{Tr}\ln O_Q + O(\hbar^2). \tag{4}$$

After removing the field-independent factor m and normalizing at $Q = 0$,

$$\Delta\Gamma_E^{(1)} = \frac{\hbar}{2}\operatorname{Tr}\ln[-\partial_\tau^2 + \Omega^2] - \frac{\hbar}{2}\operatorname{Tr}\ln[-\partial_\tau^2 + \omega^2]. \tag{5}$$

2.2 Weyl inverse through second adiabatic order

Let $A = -\partial_\tau^2 + V(\tau)$, $V = \Omega^2$, $f = p^2 + V$, and let R be the Weyl symbol of A^{-1}. The specialized Moyal equation $f \star R = 1$ is

$$f \star R = fR + \frac{\mathrm{i}}{2}(V'R_p - 2pR_\tau) - \frac{1}{8}(V''R_{pp} + 2R_{\tau\tau}) - \frac{\mathrm{i}}{48}V'''R_{ppp} + O(\partial_\tau^4). \tag{6}$$

With $R = R_0 + R_1 + R_2 + R_3 + \cdots$,

$$R_0 = \frac{1}{f}, \qquad R_1 = 0, \qquad R_2 = -\frac{V''}{2f^3} + \frac{V''p^2}{f^4} + \frac{(V')^2}{2f^4}. \tag{7}$$

The third-order terms cancel rather than being absent by assumption:

$$\{f, R_2\}_{\mathrm{PB}} = -\frac{V'''p(p^2 - V)}{(p^2+V)^4} = \frac{V'''}{24}(R_0)_{ppp} \quad\Longrightarrow\quad R_3 = 0. \tag{8}$$

Thus $R = R_0 + R_2 + O(\partial_\tau^4)$. The master integral

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}p}{2\pi}\frac{1}{(p^2+\Omega^2)^n} = \frac{\Gamma(n-\frac{1}{2})}{2\sqrt{\pi}\,\Gamma(n)}\Omega^{1-2n}, \qquad n > \tfrac{1}{2} \tag{9}$$

and $p^2/f^4 = f^{-3} - \Omega^2 f^{-4}$ give

$$G_E(\tau,\tau) = \int \frac{\mathrm{d}p}{2\pi}R = \frac{1}{2\Omega} - \frac{\ddot{\Omega}}{8\Omega^4} + \frac{3\dot{\Omega}^2}{16\Omega^5} + O(\partial_\tau^4). \tag{10}$$

2.3 Local determinant and effective coefficients

Because

$$\delta\Delta\Gamma_E^{(1)} = \frac{\hbar}{2}\int \mathrm{d}\tau\,G_E(\tau,\tau)\,\delta\Omega^2, \tag{11}$$

write, up to a total derivative,

$$L_1 = \frac{1}{2}\Omega + a\frac{\dot{\Omega}^2}{\Omega^3}, \qquad \frac{\delta}{\delta\Omega^2}\int L_1\mathrm{d}\tau = \frac{1}{4\Omega} - a\frac{\ddot{\Omega}}{\Omega^4} + \frac{3a}{2}\frac{\dot{\Omega}^2}{\Omega^5}. \tag{12}$$

Matching Eq. (10)/2 fixes $a = 1/16$. Since $2\Omega\dot{\Omega} = (\lambda/m)Q\dot{Q}$,

$$\Gamma_E^{\mathrm{loc}} = \int \mathrm{d}\tau\left[\frac{1}{2}Z(Q)\dot{Q}^2 + V_{\mathrm{eff}}(Q)\right] + O(\hbar\partial_\tau^4) + O(\hbar^2), \tag{13}$$

with

$$V_{\mathrm{eff}} = \frac{m\omega^2}{2}Q^2 + \frac{\lambda}{24}Q^4 + \frac{\hbar}{2}(\Omega - \omega), \qquad Z(Q) = m + \frac{\hbar\lambda^2 Q^2}{32m^2\Omega^5}. \tag{14}$$

3 Local real-time equation

Under $\tau = \mathrm{i}t$, the local action becomes

$$\Gamma_{\mathrm{RT}}^{\mathrm{loc}} = \int \mathrm{d}t \left[\frac{1}{2}Z(Q)\dot{Q}^2 - V_{\mathrm{eff}}(Q)\right], \qquad Z\ddot{Q} + \frac{1}{2}Z'\dot{Q}^2 + V'_{\mathrm{eff}} = 0. \tag{15}$$

Here

$$\Omega' = \frac{\lambda Q}{2m\Omega}, \quad V'_{\mathrm{eff}} = m\omega^2 Q + \frac{\lambda}{6}Q^3 + \frac{\hbar\lambda Q}{4m\Omega}, \quad Z' = \frac{\hbar\lambda^2}{32m^2}\left(\frac{2Q}{\Omega^5} - \frac{5\lambda Q^3}{2m\Omega^7}\right). \tag{16}$$

Result A - local adiabatic equation

$$\begin{aligned} 0 = &\left[m + \frac{\hbar\lambda^2 Q^2}{32m^2\Omega^5}\right]\ddot{Q} + \frac{\hbar\lambda^2}{64m^2}\left(\frac{2Q}{\Omega^5} - \frac{5\lambda Q^3}{2m\Omega^7}\right)\dot{Q}^2 \\ &+ m\omega^2 Q + \frac{\lambda}{6}Q^3 + \frac{\hbar\lambda Q}{4m\Omega}, \qquad \Omega = \sqrt{\omega^2 + \frac{\lambda Q^2}{2m}}. \end{aligned} \tag{A}$$

The truncated energy

$$E_{\mathrm{loc}} = \frac{1}{2}Z\dot{Q}^2 + V_{\mathrm{eff}}, \qquad \dot{E}_{\mathrm{loc}} = \dot{Q}\left(Z\ddot{Q} + \tfrac{1}{2}Z'\dot{Q}^2 + V'_{\mathrm{eff}}\right) = 0, \tag{17}$$

shows that the $\dot{Q}^2$ force is not friction.

Validity of Result A

A controlled derivative expansion requires a slow family $\Omega_\epsilon(t) = \bar{\Omega}(\epsilon t)$ with $\Omega_\epsilon^{(n)}/\Omega_\epsilon^{n+1} = O[(\epsilon/\Omega_{\mathrm{min}})^n]$ for all relevant n, not merely two pointwise inequalities. Analytic continuation of the Euclidean 1PI functional is an in–out construction; its causal meaning is established below by the in–in memory limit and by an independent adiabatic-mode check. Result A omits $O(\hbar\partial_t^4)$ and $O(\hbar^2)$.

4 Causal Gaussian dynamics and first-Born memory

The Heisenberg equation and $q = Q + \eta$, $Q = \langle q \rangle$, $\langle \eta \rangle = 0$ give

$$0 = m\ddot{Q} + m\omega^2 Q + \frac{\lambda}{6}Q^3 + \frac{\lambda}{2}Q\langle\eta^2\rangle_0 + \Delta_{2\mathrm{L}}, \qquad \Delta_{2\mathrm{L}} \text{ collects two-loop forces,} \tag{18}$$

because $\langle\eta^3\rangle$ starts beyond one loop. For a pure Gaussian state,

$$\hat{\eta} = \sqrt{\frac{\hbar}{2m}}\,[au + a^\dagger u^*], \qquad \ddot{u} + \Omega_Q^2 u = 0, \qquad W[u] \equiv u\dot{u}^* - \dot{u}\,u^* = 2\mathrm{i}, \tag{19}$$

$$\Omega_Q^2 = \omega^2 + \delta\Omega^2, \qquad \delta\Omega^2 = \frac{\lambda Q^2}{2m}, \qquad \langle\eta^2\rangle_0 = \frac{\hbar}{2m}|u|^2. \tag{20}$$

The Wronskian gives $[\hat{\eta}, m\dot{\hat{\eta}}] = \mathrm{i}\hbar$. With $\Delta_{2\mathrm{L}}$ omitted, the closed Gaussian system conserves

$$E_G = \frac{m}{2}\dot{Q}^2 + \frac{m\omega^2}{2}Q^2 + \frac{\lambda}{24}Q^4 + \frac{\hbar}{4}\left(|\dot{u}|^2 + \Omega_Q^2|u|^2\right). \tag{21}$$

This is reversible mean–fluctuation energy exchange, not automatic dissipation.

Choose the free-vacuum initial mode

$$u_0(t) = \frac{e^{-\mathrm{i}\omega(t-t_0)}}{\sqrt{\omega}}, \qquad \delta u(t_0) = \delta\dot{u}(t_0) = 0. \tag{22}$$

At first Born order,

$$\delta u_1(t) = -\int_{t_0}^{t} \mathrm{d}t'\, \frac{\sin[\omega(t-t')]}{\omega}\,\delta\Omega^2(t')u_0(t'), \tag{23}$$

so

$$|u(t)|^2 = \frac{1}{\omega} - \frac{1}{\omega^2}\int_{t_0}^{t} \mathrm{d}t'\, \sin[2\omega(t-t')]\delta\Omega^2(t') + R_u^{(2)}(t). \tag{24}$$

A dimensionally correct finite-time control follows from

$$\kappa_T = \sup_{t_0 \le t \le T} \frac{1}{\omega}\int_{t_0}^{t} \mathrm{d}t'\, |\delta\Omega^2(t')|. \tag{25}$$

For $\kappa_T < 1$, the Volterra–Neumann series converges and one may bound

$$|R_u^{(2)}(t)| \le \frac{\kappa_T^2(3 - 2\kappa_T)}{\omega(1-\kappa_T)^2}, \qquad t \in [t_0, T]. \tag{26}$$

Thus pointwise $|\delta\Omega^2|/\omega^2 \ll 1$ alone is not a uniform long-time guarantee.

Using Eq. (20) in Eq. (18) gives the retarded equation below, with $\Delta_{\mathrm{Born}} = (\hbar\lambda Q/4m)R_u^{(2)}$.

Result B - causal first-Born memory equation

$$
\begin{aligned}
0 &= m\ddot{Q}(t) + m\omega^2 Q(t) + \frac{\lambda}{6}Q^3(t) + \frac{\hbar\lambda}{4m\omega}Q(t) \\
&\quad - \frac{\hbar\lambda^2}{8m^2\omega^2}Q(t)\int_{t_0}^{t} \mathrm{d}t' \, \sin[2\omega(t-t')]Q^2(t') + \Delta_{\mathrm{Born}}(t) + \Delta_{2\mathrm{L}}(t).
\end{aligned} \tag{B}
$$

Finite-t_0 oscillations are state-preparation transients. They describe a real quench when that is the preparation, but are artifacts if the intended state is the interacting adiabatic vacuum. For comparison with the Euclidean vacuum functional, take $t_0 \to -\infty$ with $e^{-\epsilon(t-t')}$ inserted before $\epsilon \to 0^+$.

5 Local limit and independent coefficient checks

5.1 Low-frequency expansion of the memory kernel

With $a = 2\omega$, define

$$
I[f](t) = \lim_{\epsilon\to 0^+}\int_0^{\infty} \mathrm{d}s \, e^{-\epsilon s}\sin(as) f(t-s). \tag{27}
$$

For $f(t) = \int(\mathrm{d}\nu/2\pi)e^{-\mathrm{i}\nu t}\tilde{f}(\nu)$,

$$
\widetilde{I[f]}(\nu) = \frac{a}{a^2 - (\nu + \mathrm{i}0)^2}\tilde{f}(\nu). \tag{28}
$$

If the spectrum is confined well inside $|\nu| < a$, then

$$
I[f] = \frac{f}{a} - \frac{\ddot{f}}{a^3} + \frac{f^{(4)}}{a^5} + \cdots. \tag{29}
$$

The nearest poles are at $\nu = \pm 2\omega$; no local series is controlled near that pair resonance. For $f = Q^2$,

$$
-\frac{\hbar\lambda^2}{8m^2\omega^2}QI[Q^2] = -\frac{\hbar\lambda^2 Q^3}{16m^2\omega^3} + \frac{\hbar\lambda^2}{32m^2\omega^5}\left(Q^2\ddot{Q} + Q\dot{Q}^2\right) + O(\hbar\partial_t^4). \tag{30}
$$

Introduce the dimensionless amplitude

$$
x = \frac{\lambda Q^2}{2m\omega^2}, \qquad \frac{1}{\Omega} = \frac{1}{\omega}\left(1 - \frac{x}{2} + O(x^2)\right), \qquad \frac{1}{\Omega^5} = \frac{1}{\omega^5}\left(1 - \frac{5x}{2} + O(x^2)\right). \tag{31}
$$

Expanding Result A through $O(\hbar\lambda^2)$ and second derivative order gives

$$
\begin{aligned}
0 &= m\ddot{Q} + m\omega^2 Q + \frac{\lambda}{6}Q^3 + \frac{\hbar\lambda Q}{4m\omega} - \frac{\hbar\lambda^2 Q^3}{16m^2\omega^3} \\
&\quad + \frac{\hbar\lambda^2}{32m^2\omega^5}\left(Q^2\ddot{Q} + Q\dot{Q}^2\right) + \text{higher orders},
\end{aligned} \tag{32}
$$

which exactly matches the displayed local expansion of Result B at that order.

5.2 Real-time adiabatic/WKB check

The stronger check does not expand in x. For the adiabatic vacuum, take

$$
u(t) = \frac{1}{\sqrt{W(t)}}\exp\left[-\mathrm{i}\int^{t} W(t')\mathrm{d}t'\right], \qquad W^2 = \Omega^2 - \frac{\ddot{W}}{2W} + \frac{3\dot{W}^2}{4W^2}. \tag{33}
$$

Iterating through second adiabatic order,

$$
|u|^2 = \frac{1}{\Omega} + \frac{\ddot{\Omega}}{4\Omega^4} - \frac{3\dot{\Omega}^2}{8\Omega^5} + O(\partial_t^4). \tag{34}
$$

Using

$$
\dot{\Omega} = \frac{\lambda Q\dot{Q}}{2m\Omega}, \qquad \ddot{\Omega} = \frac{\lambda(\dot{Q}^2 + Q\ddot{Q})}{2m\Omega} - \frac{\lambda^2 Q^2\dot{Q}^2}{4m^2\Omega^3}, \tag{35}
$$

the tadpole force becomes

$$
\frac{\lambda}{2}Q\langle\eta^2\rangle_0 = \frac{\hbar\lambda Q}{4m\Omega} + \frac{\hbar\lambda^2 Q^2}{32m^2\Omega^5}\ddot{Q} + \frac{\hbar\lambda^2}{64m^2}\left(\frac{2Q}{\Omega^5} - \frac{5\lambda Q^3}{2m\Omega^7}\right)\dot{Q}^2 + O(\hbar\partial_t^4), \tag{36}
$$

reproducing every displayed one-loop coefficient in Result A.